

Certifying a compiled circuit from its own garbage

Track B: the anti-controlled Hadamard test, upgraded with classical shadows

Garbage Collectors

TBD-NAME TBD-NAME TBD-NAME

Qiskit Hackathon 2026 — Topic 5

after Faehrman, Eisert & Kueng, *PRL* **135**, 150603 (2025)

14 August 2026

Outline

The question

The instrument

Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

Every number carries the `RESULTS.md` row that produced it, shown as `[R0xx]`. Where a baseline beats us, the slide says so.

The question

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What we claim

You compiled U into something cheaper. Is it still right?

Every practical quantum simulation replaces the target evolution $U(t)$ with a cheaper approximation W — a Trotter product formula, a transpiler setting, or an **AQC-compressed ansatz**.

The question that decides whether the result means anything:

how wrong is W , and wrong in *what*?

Classically you answer it by simulating both. That is exactly the thing you cannot do at the sizes where you need the answer.

Track B answers it *on the device*: put W on one ancilla branch and U on the other, and the interference is the overlap of two dynamics.

W = what you can afford to run

U = what you meant to run

$$\chi_{AB}(t) = \langle \psi | W^\dagger U | \psi \rangle$$

measured, not simulated

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Track B: anti-controlled Hadamard test

Replace controlled- U with “ W when the ancilla is $|0\rangle$, U when it is $|1\rangle$ ”. Then, with the *same* quadrature convention as the ordinary test,

$$\langle Z_a \rangle_\phi = \operatorname{Re} [e^{i\phi} \langle \psi | W^\dagger U | \psi \rangle], \quad \phi = 0 \rightarrow \operatorname{Re} \chi_{AB}, \quad \phi = -\frac{\pi}{2} \rightarrow \operatorname{Im} \chi_{AB}$$

Setting $W = \mathbb{I}$ recovers the ordinary Hadamard test exactly, so this is a strict generalisation — which also means it needs *its own* validation, not inherited trust.

statevector identity, frozen 3-site (3 times $\times$ 2 reps $\times$ 2 quadratures)	8.99×10^{-15} [R034]
statevector identity, 2-site (4 times $\times$ 2 reps $\times$ 2 quadratures)	4.56×10^{-15} [R042]
$\chi_{AB}(0) = 1$ exactly, as it must be (both branches are the identity)	yes

A bug this caught. Our first reference took the $|1\rangle$ -control block as the *upper half* of the gate matrix. The notebook builds the gate with the ancilla as its *lowest* bit, so the block is the odd sublattice. The wrong version passes at $t = 0$ — where both branches are the identity — and fails everywhere else. It was caught before any QPU time was spent.

What the shadows add: delete the final Hadamard

Before the final Hadamard the state is $(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle)/\sqrt{2}$. Tracing the ancilla out of the *un-Hadamarded* circuit gives two different things at once:

$$\mathrm{tr}_a[\rho_{\mathrm{out}}] = \frac{1}{2}(W\rho W^\dagger + U\rho U^\dagger), \quad \mathrm{tr}_a[(Z \otimes \mathbb{I})\rho_{\mathrm{out}}] = \frac{1}{2}(W\rho W^\dagger - U\rho U^\dagger)$$

Both are *independent of ϕ* , so one quadrature suffices. Add and subtract:

$\langle O \rangle_W$ and $\langle O \rangle_U$ separately, for every Pauli O , from one family of circuits.

That is the *per-observable disagreement profile* — the part of Track B the scaffold flags and that our own R034 explicitly did **not** deliver. Validated to 4.44×10^{-15} across 4 times $\times$ 6 observables before submission. [R042]

The ancilla says *how much* W is wrong. The garbage says *which observable*.

The question

The instrument

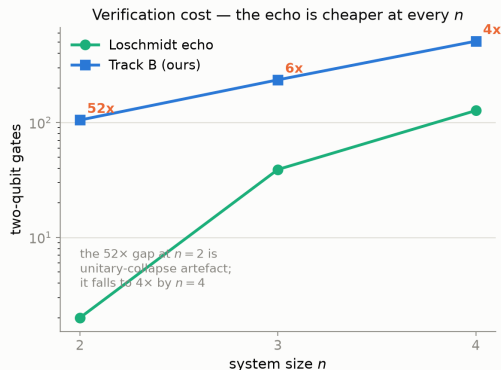
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We are not the only way to do this — and not the cheapest



What each method actually returns

gives a number	yes	yes	yes	yes	yes
runs on a QPU	--	yes	yes	yes	yes
keeps the phase	--	--	--	yes	yes
says WHICH observable broke	--	--	--	--	yes
	classical fidelity	Loschmidt echo	Hilbert-Schmidt	Track B ancilla only	Track B + shadows

Loschmidt echo: prepare, apply U , apply $W^\dagger$, un-prepare, read the all-zeros probability.

Hilbert–Schmidt test (Khatri *et al.*, *Quantum* 3, 140 (2019)): Bell pairs on $2n$ qubits give the *state-independent* $|\text{Tr}[W^\dagger U]/d|^2$. Both verified against exact values before being used as baselines.

[R043]

The honest scoreboard

	qubits	2q gates	phase?	state-indep.?	which observable?
classical fidelity (AQC's own score)	—	—	n/a	yes	no
Loschmidt echo	n	2	no	no	no
Hilbert–Schmidt test	$2n$	16	no	yes	no
Track B, ancilla only	$n+1$	136	yes	no	no
Track B + shadows (ours)	$n+1$	136	yes	no	yes

- **The echo is cheaper than us at every n we measured** — $6.0\times$ at $n=3$ and $4.0\times$ at $n=4$. If all you want is a scalar “how close is W to U ”, *use the echo*. [R043]
- The $52\times$ gap at $n=2$ is a *unitary-collapse artefact* (any product of 2-qubit unitaries fuses into one). We quote $4\text{--}6\times$, never $52\times$.
- The HST answers a *stronger* question than either overlap, and costs $2n$ qubits for it.

Our arm has to be justified by what it returns, never by being cheaper.

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AQC compression, certified by our own Track-B observable

Approximate Quantum Compiling (qiskit-addon-aqc-tensor, quimb+jax MPS backend) optimises a shallow ansatz against an MPS target. It scores itself with a *classical* fidelity — the one number you lose access to at scale.

Set W = the AQC circuit and U = exact: then $\chi_{AB} = \langle \psi | W^\dagger U | \psi \rangle$ **is the compression fidelity**, measured from shots.

	Trotter	exact synth.	AQC
<i>uncontrolled</i> $n=3$ (infid. 1.4×10^{-4})	68	19	9
<i>uncontrolled</i> $n=4$ (infid. 2.6×10^{-4})	104	95	15
<i>controlled</i> $n=3$ — what a Hadamard test runs	—	50	132
<i>controlled</i> $n=4$	—	236	216

The compressor and the certificate come from two different halves of this project. **But the story reverses under control:** at $n=3$ AQC is $2.6\times$ worse, because controlling an ansatz pays overhead on every gate while the direct block construction pays none. Crossover sits at $n \sim 4$. [R035]

The question

The instrument

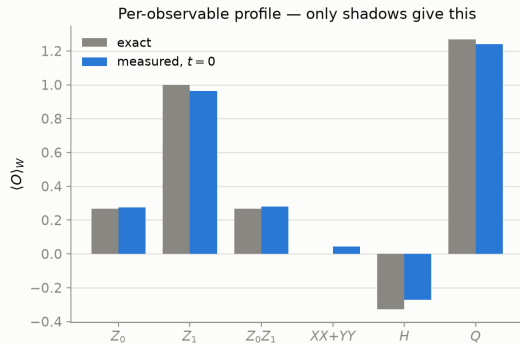
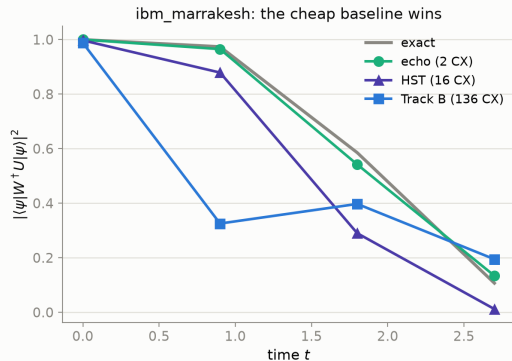
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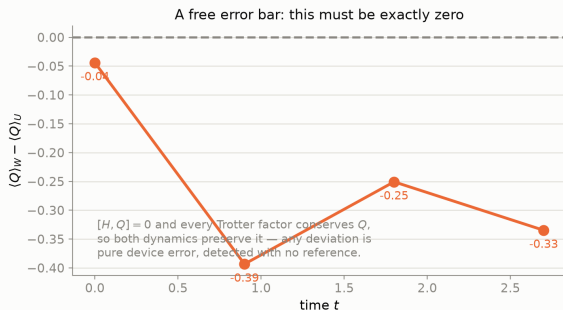
What we claim

First Track B on a QPU — and the cheap baseline beat us



ibm_marrakesh, 116 circuits $\times$ 1000 shots, all four arms in *one* job so the comparison is same-device same-hour. Predictions were recorded before submission. **The 136-CX Track-B arm degrades fast:** clean at $t = 0$ (errors all ≤ 0.053), noise dominated by $t = 0.9$. The 2-CX echo tracks the exact curve to ~ 0.04 throughout. [R044]

The garbage also ships a free error bar



$Q = \sum_j Z_j$ commutes with H , and every Trotter factor conserves it — so $\langle Q \rangle_W - \langle Q \rangle_U$ is **exactly zero** for any W in this family.

Measured: $-0.04, -0.39, -0.25, -0.34$.

[R044]

That deviation is **pure device error, detected with no reference state, no exact diagonalisation and no simulation** — and it correctly flags the same times where the other observables went bad.

It is a diagnostic, not a correction: we do *not* post-select on it, which would bias the shadow estimators.

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What we claim, and what we do not

We claim

- Track B **completed**: the per-observable disagreement profile via the X-basis readout, which our own R034 left undone. Identity validated to 4×10^{-15} on two models. [R042]
- An AQC-compressed circuit **certified by a shot-based observable** rather than a classical fidelity — the certificate survives to sizes where the classical score does not. [R035]
- Track B run on a **QPU for the first time in this project**, against two literature baselines in the same job. [R044]

We do not claim

- **That our method is the best way to get the scalar overlap.** It is not — the Loschmidt echo is $4\text{--}6\times$ cheaper and, on hardware, markedly more accurate.
- **That AQC helps a Hadamard test at these sizes.** Under control it is $2.6\times$ worse at $n=3$; the crossover is at $n \sim 4$. [R035]
- Any advantage over classical computation. The instance is 2–3 qubits and a laptop solves it exactly — which is the point: every number here is graded against truth.

Reproduce it

- `evidence/scripts/track_b_ab_tester.py` — Track B derivation + statevector validation + simulator study [R034]
- `evidence/scripts/track_b_baselines.py` — Loschmidt echo, Hilbert–Schmidt test, and the $n=2, 3, 4$ cost scaling [R043]
- `evidence/scripts/track_b_hardware_submit.py` — validate-then-submit, all four arms; refuses to submit if the identity check fails [R042]
- `evidence/scripts/track_b_hardware_fetch.py` — the analysis behind every hardware number on these slides [R044]
- `evidence/scripts/aqc_compression.py` — AQC + Track-B certification [R035]

Garbage Collectors

Thank you — questions welcome.